

device drivers. If the problem persists, uninstall problematic devices through Device Manager and all their associated software, and restart the computer. Windows XP will auto-detect the hardware and ask you to provide the location for the software drivers. Specify the location of the new drivers and let Windows complete the installation. Some devices may require a firmware update. See the *Updating the BIOS* section of this guide to understand the general principles on how it is done. Refer to the manufacturer's Web site for details on the exact steps to be followed.

► **Problem:** A BSOD during shutdown.

Cause: This is most likely a problem with the hardware; it may also be caused by rogue software or problems with system services.

Solution: Note down the stop message error code and the friendly name. Refer to the *Troubleshooting BSODs* section for general principles on how to troubleshoot the problem. Troubleshoot problems with Windows system services as explained earlier in this chapter.

► **Problem:** Shutdown is very slow.

Cause: Windows is waiting for all running programs and services to stop before shutting down.

Solution: In XP, many programs (including Windows itself) install services that run in the background, waiting for some action from the user. Some of these may take a long time to shut themselves down after receiving the Shutdown command from Windows. Identifying the service or services causing this problem will mostly be a trial and error process. If you have installed some software recently it might have a service that is slow in exiting and is delaying the shutdown process.

To find out the list of services that are running on your system, click Start > Run, type in "services.msc" and click OK. This launches the Services Manager. Look through the list and identify any services related to software you have recently installed.